

ACHINTYA RAI

925-202-1555 | achintyaraism@gmail.com | linkedin.com/in/achintya-rai | github.com/Azurinine

EDUCATION

University of California, San Diego
Bachelor of Science in Computer Engineering

Sept. 2024 – June 2028
GPA: 3.9/4.0

WORK EXPERIENCE

Adobe
Software Engineering Intern

San Jose, CA
June 2026 – Sept. 2026

- Profiled **60.6M+** payment events in Databricks to establish the canonical data source for downstream monitoring, surfacing **86,387** duplicate records and a **47%** null rate across timestamp fields.
- Diagnosed a **26-hour** outage via hourly volume analysis across **1,272** monitoring segments.
- Built **SQL** data pipelines and an **MLOps** workflow for weekly retraining and hourly inference.

VALT Health
Founding Software Engineer

Remote
Aug. 2025 – Feb. 2026

- Engineered a **Flask** backend ingesting **100k+** datapoints from fitness devices (Fitbit, Apple Watch), designing database schemas to unify data formats across **20+** device sources.
- Implemented async batch processing in **GCP**, boosting speed **35%** and cutting redundant writes **60%**.
- Automated environment setup and data seeding via **Bash** scripts, cutting testing time **80%**.

Decklar (formerly Roambee)
Software Engineering Intern

Sunnyvale, CA
June 2024 – Aug. 2024

- Developed real-time shipment monitoring modules for **Amazon, Coca-Cola, and Samsung**, optimizing **Python/NumPy** data pipelines to sustain a **15%** capacity boost across **1M+** weekly sensor events.
- Implemented advanced data structures for analytics modules, accelerating reporting speeds **20%**.
- Refined a Panels-based **UI/UX** through iterative user testing, improving workflow efficiency **15%**.

RESEARCH

Graph Theory & Neural Architecture
Undergraduate Research Fellow (NSF REU)

San Diego, CA
March 2026 – Present

- Engineered a **Python** pipeline for **LLM** embedding analysis, building weight-matrix extraction utilities that shrank data footprints from **500GB** checkpoints to **13GB** localized manifolds.
- Optimized preprocessing for a **30x** runtime cut, enabling analysis on **8GB RAM** systems.
- Applied **Louvain Clustering** and Rent's Rule to raise Inter-Cluster density from **0.18%** to **13.25%**.

PROJECTS

SquaredUpChess | *React, TypeScript, Django, WebSockets, AWS, Docker, CI/CD* 

Aug. 2025 – Dec. 2025

- Engineered a full-stack **AWS** platform for scheduling, real-time tracking, and payments.
- Built a responsive **React/TypeScript** UI with **WebSockets** for low-latency real-time updates.
- Architected **20+ Django REST API** endpoints for authentication, sessions, and transactions.
- Established a **CI/CD** pipeline with automated testing for reliable **Docker** deployments.

MovieRecapinator | *Java, Spring Boot, Maven, JavaScript, IMDb API* 

April 2025

- Developed a recommendation engine in **Java/Spring Boot** processing **700,000+** IMDb API entries.
- Delivered a full-stack MVP with a 4-person team at **DataHacks** within a **24-hour** sprint.

TECHNICAL SKILLS

Languages: Python, C/C++, TypeScript, Java, JavaScript, SQL (Postgres/SQLite), Bash, HTML/CSS

Development: React, Django, Node.js, FastAPI, Flask, Asyncio, WebSockets, REST APIs, JUnit, PyTest

AI & Cloud: PyTorch, TensorFlow, AWS, GCP, Docker, CI/CD, GitHub Actions, Gemini/OpenAI API, MCP, Claude Code, Linux